Review article

Early versus delayed feeding after endoscopic submucosal dissection: a systematic review and meta-analysis (protocol)

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Short running title: Early vs. delayed feeding after ESD

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1. Introduction

Gastric cancer is one of the most common cancer and the third leading cause of cancer death worldwide; annually, approximately one million people are diagnosed with gastric cancer in the world, of whom around 782,000 died [1, 2]. Upper gastrointestinal endoscopy has become widely used for the detection and early treatment of gastric cancer [3]. Endoscopic submucosal dissection (ESD) is an established resection method and is being practiced removing early cancers and large lesions from the gastrointestinal tract [4].

ESD had a low risk of perforation [5, 6, 7], short hospital stays [8, 9, 10], higher quality of life (QOL) [10, 11, 12], while there are no significant differences in overall survival compared to gastrectomy for early gastric cancer [10, 13, 14]. Due to its safety and effectiveness, ESD has increasingly performed worldwide [15] With regard to ESD, quality of care, such as hospital stay and QOL, is also important [16]. The timing of feeding would contribute to such a care [17, 18]. The early and delayed feeding was defined as the initiation of feeding after ESD within 1 day and after 2 days, respectively [19, 20], while the evidence of the timing of feeding on quality of care has not been established due to lack of meta-analysis and guidelines recommendation [21].

Therefore, the present study will aime to assess the quality of care of early versus delayed feeding after ESD.

2. Methods

We followed the Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 for preparing this protocol [22]. We will publish this protocol in protocols.io (https://www.protocols.io/)

Inclusion criteria of the articles for review

Randomized controlled trials (RCTs) will be included to assess the effecticacy and safety of early versus delayed feeding after ESD. All studies will be included, reported as full text, those published as abstract only, and unpublished data, regardless of language or country restrictions. Inclusion criteria will be adult patients (≥ 18 years of age) who underwent ESD for gastric lesions. Exclusion criteria will be residual and recurrent lesions after ESD.

Type of outcomes

Primary outcomes will be post-ESD bleeding, patients' satisfaction, and lengths of hospital stay (days). Post-ESD bleeding will be defined as clinical evidence of bleeding after ESD, as represented by hematemesis, melena, or hematochezia after normalization of stool color, or a decrease in hemoglobin levels of ≥2.0 g/dL, after consecutive stable hemoglobin level and/or active bleeding confirmed by endoscopic evaluation. Patients' satisfaction will be defined as the mean score of the numeric rating scale for allover patients' satisfaction after ESD. The lowest score corresponds to no very dissatisfaction while the highest score corresponds to very satisfaction. Secondary outcomes will be abdominal pains and post-ESD ulcer healing status, and all adverse events. Abdominal pain will be defined as the mean score of the numeric rating scale for patients' abdominal pain after ESD. The lowest score corresponds to no pain while the highest score corresponds to unbearable pain. Post-ESD ulcer healing status will be defined as proportion of patients with ulcer stage S1 and S2 by follow-up esophagogastroduodenoscopy at 2 months after ESD. All adverse events will be defined by the original articles.

Search method

The following databases will be searched: the Cochrane Central Resister of Controlled Trials (CENTRAL), MEDLINE via Ovid, and EMBASE via ProQuest Dialog (Appendix 1). The following databases will be also searched for ongoing or unpublished trials: the World Health Organization International Clinical Trials Platform Search Portal (ICTRP), and ClinicalTrials.gov (Appendix 2). The reference lists will be checked for studies, including international guidelines published by the American Gastroenterological Association, the European Society of Gastrointestinal Endoscopy, and Japan Gastroenterological Endoscopy Society [4, 11, 15], as well as the reference lists of eligible studies and articles citing eligible studies. The original authors for unpublished or additional data will be asked if necessary.

Data collection and analysis

Two reviewers (Jun W and Joji W) will independently screen titles and abstracts, followed by the assessment of the eligibility based on the full texts. If relevant data is missing, the original authors will be contacted. Disagreements between the two reviewers will be resolved by discussion, and if this failed, a third reviewer (KK) acted as an arbiter.

Two reviewers (Jun W and Joji W) will independently perform data extraction of the included studies using standardized data collection form, including the information on study design, study population, patients' characteristics, primary and secondary outcomes. The risk of bias will be independently evaluated using the Risk of Bias 2 [23]. Any disagreements will be resolved by discussion, and if failed, a third reviewer (KK) acted as an arbiter, if necessary.

We will pool the relative risk ratios and the 95% confidence intervals (CIs) for the following binary variables: post-ESD bleeding and post-ESD ulcer healing status. We will pool the mean differences (MD) and the 95% CIs for the lengths of hospital stay. We will pool the standard mean differences (SMD) and the 95% CIs for patients' satisfaction and abdominal pains. If some scales increase with patients' satisfaction while others decreased, it multiplies the mean values from one set of studies by -1 to ensure that all the scales point in the same direction. The intention-to-treat (ITT) analysis will be performed for all dichotomous data as much as possible. For, continuous data, missing data will not impute based on the recommendation by Cochrane handbook [24]. Meta-analysis will be performed about the available data in the original study.

Missing outcomes

When original studies only report standard error or p value, the standard deviation will be calculated based on the method by Altman [25]. If these values is missing when the original author was contacted, the standard deviation will be calculated by the CI and the t value based on the method by the Cochrane handbook [24], or validated method [26]. Validity of these methods will be analyzed by sensitivity analysis.

Assessment of heterogeneity

The heterogeneity will be evaluated by visual inspection of the forest plots and calculating the I^2 statistic (I^2 values of 0% to 40%: might not be important; 30% to 60%: may represent moderate heterogeneity; 75% to 100%: considerable heterogeneity) [24]. When heterogeneity is identified ($I^2 > 50\%$), the reason of heterogeneity will be assessed using subgroup analysis.

Assessment of reporting bias

When more than 10 trials are pooled, a funnel plot will be created and examined to explore possible publication biases. Egger test will be performed as well. When less than 10 trials, the test will be not performed [24].

Meta-analysis

Meta-analysis will be performed using Review Manager software (The Cochrane Collaboration, Copenhagen, Denmark) and will be used a random-effects model.

Subgroup analysis

We will evaluate the subgroup analyses of the primary outcomes in patients with lesion size (< 20 mm versus ≥ 20 mm) when sufficient data were available.

Sensitivity analysis

The following sensitivity analyses of the primary outcomes will be undertaken: exclusion of studies using imputed statistics, and missing participants: verify the robustness of the results by seeking informative missingness odds ratios [26].

Summary of findings table

Summary of finding table will be made for the primary and secondary outcome based on the Cochrane handbook [24]. The study quality will be assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach [27].

3. Conflict of interest

The authors declare no conflicts of interests.

4. References

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Appendix 1

CENTRAL search strategy

#1. MeSH descriptor: [Stomach Neoplasms] explode all trees

#2. (gastr$ or stomach):ab,ti,kw

#3. (polyp$ or neoplas$ or tumour$ or tumor$ or adenom$ or lesion$ or carcinom$ or adenocarcinom$ or cancer$):ab,ti,kw

#4. #2 AND #3

#5. #1 OR #4

#6. MeSH descriptor: [Endoscopic Mucosal Resection] explode all trees

#7. ((endoscopic mucosal resection) OR (endoscopic submucosal dissection) OR (ESD) OR (EMR)):ab,ti,kw

#8. #6 OR #7

#9. #5 AND #8

#10. MeSH descriptor: [Diet] explode all trees

#11. ((feed$) OR (fast$) OR (diet$) OR (intake$)):ab,ti,kw

#12. #10 OR #11

#13. #9 AND #12

MEDLINE via Ovid search strategy

1. exp stomach neoplasms/
2. (gastr$ or stomach).tw.
3. (polyp$ or neoplas$ or tumour$ or tumor$ or adenom$ or lesion$ or carcinom$ or adenocarcinom$ or cancer$).tw.
4. 2 and 3
5. 1 or 4
6. exp endoscopic mucosal resection/
7. (endoscopic mucosal resection OR endoscopic submucosal dissection OR EMR OR ESD).tw.
8. 6 or 7
9. 5 and 8
10. exp diet/
11. ((feed$) OR (fast$) OR (diet$) OR (intake$)).tw.
12. 10 or 11
13. 9 and 12

EMBASE via ProQuest Dialog search strategy

S1 (EMB.EXACT.EXPLODE("stomach cancer"))

S2 (ab(gastric) OR ti(gastric) OR ab(stomach) OR ti(stomach))

S3 (ab(polyp) OR ti(polyp) OR ab(polyps) OR ti(polyps) OR ab(neoplasm) OR ti(neioplasm) OR ab(neoplasms) OR ti(neioplasms) OR ab(tumour) OR ti(tumour) OR ab(tumours) OR ti(tumours) OR ab(tumor) OR ti(tumor) OR ab(tumors) OR ti(tumors) OR ab(adenoma) OR ti(adenoma) OR ab(adenomas) OR ti(adenomas) OR ab(lesion) OR ti(lesion) OR ab(lesions) OR ti(lesions) OR ab(carcinoma) OR ti(carcinoma) ab(carcinomas) OR ti(carcinomas) OR ab(adenocarcinoma) OR ti(adenocarcinoma) OR ab(adenocarcinomas) OR ti(adenocarcinomas) OR ab(cancer) OR ti(cancer) OR ab(cancers) OR ti(cancers))

S4 S2 AND S3

S5 S1 OR S4

S6 (EMB.EXACT.EXPLODE("endoscopic mucosal resection"))

S7 (ab(endoscopic mucosal resection) OR ti(endoscopic mucosal resection) OR ab(endoscopic submucosal dissection) OR ti(endoscopic submucosal dissection) OR ab(ESD) OR ti(ESD) OR ab(EMR) OR ti(EMR))

S8 S6 OR S7

S9 S5 AND S8

S10 (EMB.EXACT.EXPLODE("dietary intake"))

S11 (ab(feed) OR ti(feed) OR ab(feeding) OR ti(feeding) OR ab(fast) OR ti(fast) OR ab(fasting) OR ti(fasting) OR ab(diet) OR ti(diet) OR ab(intake) OR ti(intake))

S12 S10 OR S11

S13 S9 AND S12

Appendix 2

ICTRP search strategy

(“endoscopic mucosal resection” OR “endoscopic submucosal dissection” OR “EMR” OR “ESD”) AND (“feed” OR “fast” OR “diet” OR “intake”)

ClinicalTrials.gov search strategy

Other terms: (“endoscopic mucosal resection” OR “endoscopic submucosal dissection” OR “EMR” OR “ESD”) AND ("feed" OR "fast" OR "diet" OR "intake")